# Polarization-Aware Modular IR Apertures for Interference Mitigation in Multi-User Indoor Networks

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***Abstract*—A polarization-aware method improves modular clustered IR optical apertures to mitigate inter-user interference in multi-user near-field indoor networks. Assigning orthogonal polarization with aligned receivers enhances SINR and leads to a 77% reduction in average BER, confirming improved reliability.**

**Keywords—Indoor optical wireless networks, near-field, multi-beam focusing, phased array, polarization**

## I. Introduction

Indoor optical wireless networks represent a compelling option for high-capacity, low-latency, and more secure communication in next-generation wireless systems. In particular, Infrared (IR) beam focusing stands out for its ability to serve spatially separated users with high-directional beams [1]. To meet the demands of scalable multi-user beam focusing, we recently introduced a modularly clustered IR optical aperture architecture with sub-cluster segmentation for simultaneously generating multiple near-field beams at designated receiver locations and a dual-carrier strategy for suppressing grating lobe formation [2]. By utilizing spatial resolution, the proposed method outperforms optical phased arrays, micro-electro-mechanical systems, and spatial light modulator-based beam steering systems, which are limited by angular resolution. Despite these improvements, inter-user interference at receiver locations remained considerable, and it continued to limit the achievable signal-to-interference-plus-noise ratio (SINR) and overall system reliability. The previous system neglected polarization effects based on the assumption of polarization-insensitive reception.

This work presents a polarization-aware enhancement for the modular-clustered IR optical aperture framework. Strategic polarization assignment to radiative sub-clusters and detection alignment at the receiver effectively mitigate inter-user interference. Simulation results demonstrate substantial improvements in SINR and bit error rate (BER) performance, offering a simple yet effective extension to prior architecture.

## II. System Model and Proposed Method

### A. System Overview

The proposed system refines upon the modularly clustered IR optical aperture architecture introduced in our previous work [2]-[4]. The aperture is placed centrally on a planar ceiling,

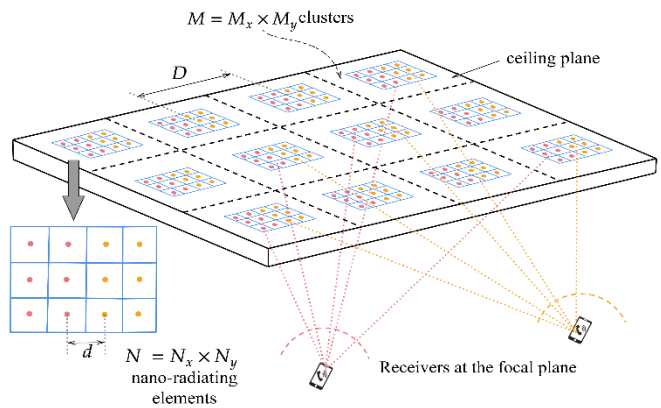


Fig. 1. Proposed optical aperture on a uniform rectangular ceiling plane representing sub-cluster segmentation for two receivers.

which consists of a planar array of main clusters arranged across $M = M_x \times M_y$ positions with inter-cluster spacing $D$. Each main cluster contains $N = N_x \times N_y$ IR elements with inter-element spacing $d$. The distance $D$ lies in the meter range while the $d$ lies in the nanometer range. Each element integrates a phase shifter and attenuator, enabling individual phase and amplitude control for focusing. Once $Q$ active receiver positions ($Q \leq N$) are identified, we conduct a systematic cluster segmentation process that dynamically arranges uniformly sized sub-clusters inside main clusters. Each receiver acquires one sub-cluster from each main cluster, enabling independent beam control via phase delay compensation. We deploy a dual-carrier transmission scheme to suppress grating lobes formation, utilizing two slightly offset IR wavelengths, $\lambda_d$ (data carrier) and $\lambda_b$ (reference carrier) simultaneously. The presence of $\lambda_b$ disrupts reinforcement of grating lobes caused by larger $D$ distance, thereby enhancing spatial resolution across the near-field focal plane. A schematic illustration of the proposed aperture for two receivers is shown in Fig. 1. Further technical details regarding the sub-cluster segmentation algorithm and dual-carrier design principles can be found in [2].

### B. Polarization-Aware Beam Focusing

Although the modular clustered optical aperture attains multi-user near-field focusing, inter-user interference can remain as a significant limitation, specially for closely placed receivers.

Funding for this work was provided by the Australian Research Council (ARC) under Discovery Project DP230101757 and Monash International Postgraduate Research Scholarship.

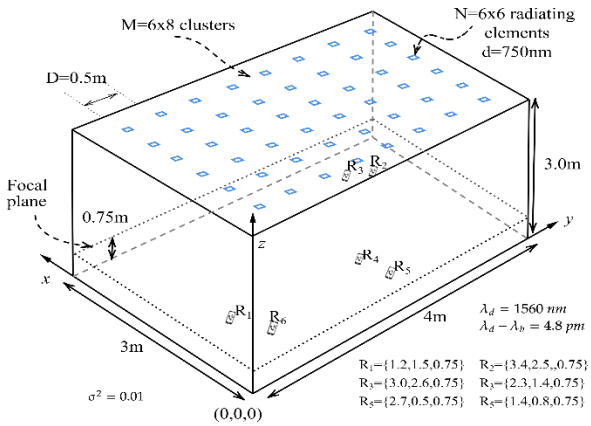


Fig. 2. Configuration of cluster aperture on a planar ceiling with six receivers

While prior designs in [2]-[4] treated polarization effects as negligible, we incorporate polarization vectors matching at both the optical aperture and receiver end in this work. This leverages selective interference mitigation through polarization orthogonality without altering the aperture configuration.

We assign a unit polarization vector $\hat{\vartheta}_i$ to each radiative sub-cluster according to its target receiver, chosen from a predefined set of orthogonal polarization states. Each receiver configures its detection vector $\hat{d}_q$ to match the polarization of the desired beam. At $q^{\text{th}}$ receiver location $\boldsymbol{r}_q$, the total electric field denoted as

$$\vec{E}(\boldsymbol{r}_q) = \sum_{i=1}^{Q} \psi_i(\boldsymbol{r}_q)\, \hat{\vartheta}_i, \tag{1}$$

where $\psi_i(\boldsymbol{r}_q)$ represents the beam contribution of the sub-cluster group associated with $i^{\text{th}}$ receiver. Such a sub-cluster group is formed by comprising one sub-cluster from each main cluster across the aperture. Then, detected field at the $q^{\text{th}}$ receiver is given by

$$E_{det}^{q} = \hat{d}_q \cdot \vec{E}(\boldsymbol{r}_q). \tag{2}$$

This formulation enables selective interference suppression at the $q^{\text{th}}$ receiver by restrictively aligning the assigned sub-cluster group polarization vector and the receiver detection axis based on polarization orthogonality.

### C. SINR Modeling

Following polarization-aware field detection, we consider temporal signal modeling at each receiver based on the detection field to evaluate communication performance. Four-level pulse amplitude modulation (PAM-4) using symbol levels {-3, -1, 1, 3} is adopted, ensuring normalization to unit average power for SINR calculation. The receiver incorporates additive white Gaussian noise (AWGN) to represent thermal and background noise effects. The SINR at the $q^{\text{th}}$ receiver is expressed as

$$SINR_q = \frac{\left|E_{det}^{q}\right|^2}{\sum_{i=1, i\neq q}^{Q} \psi_i(r_q)(\hat{d}_q \cdot \hat{\vartheta}_i) + \sigma^2}\ , \tag{3}$$

where $\sigma^2$ denotes the noise variance. This SINR definition captures the combined effects of spatial interference, polarization mismatch, and receiver noise, delivering a basis for quantitative performance evaluation in simulations.

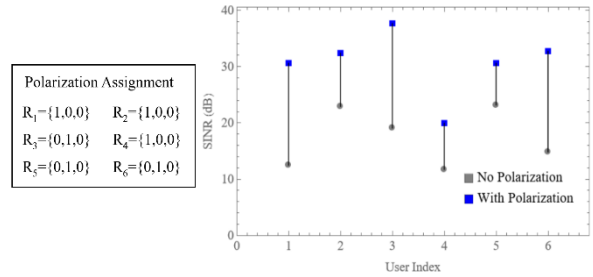


Fig. 3. Optimal Polarization assignment and corresponding SINR performance for six users with and without polarization awareness.

## III. SIMULATION RESULTS

The indoor setup, featuring an optical aperture with six no of receiver coordinates and dual-carrier wavelengths, is shown in Fig. 2. With two orthogonal polarization vector states ({1,0,0}, {0,1,0}), brute-force search is applied to find optimal polarization assignment for six receivers which results in best SINR values. Heuristic approaches such as simulated annealing are applicable for larger receiver counts to obtain near-optimal configurations. As shown in Fig. 3, the optimized polarization assignment yields notable SINR gains across all six users, with improvements over 15dB in several cases compared to the polarization-insensitive reference. This enhancement translates to a 77% reduction in average BER, highlighting practical communication benefits.

## IV. CONCLUSION

A polarization-aware scheme for modularly clustered IR apertures reduces inter-user interference, achieving major SINR improvements and a 77% BER reduction without modifying the aperture structure.


## ACKNOWLEDGMENT

This research is funded by the Australian Research Council (DP230101757). S.A. acknowledges the support of the Monash International Postgraduate Research Scholarship and AχL team at Monash University.


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